



Practice Task + Quiz

(Recap of Week 2)

 **Day 14**

1. What is the difference between = and == in JavaScript?

2. What's the output?

```
let a = 10;  
a += 5;  
console.log(a);
```

3. Which operators are used to increment and decrement a variable?

4. What is the difference between == and ===?

5. What's the result?

```
5 == '5'  
5 === '5'
```

6. Check :

```
let age = 18;  
if (age >= 18) {  
  console.log("Eligible to vote");  
}
```

7. What do &&, ||, and ! mean?

8. Example :

```
let isLoggedIn = true;
let isAdmin = false;

if (isLoggedIn && isAdmin) {
  console.log("Welcome, Admin");
} else {
  console.log("Access Denied");
}
```

9. Example:

```
let marks = 85;

if (marks >= 90) {
  console.log("Grade A+");
} else if (marks >= 75) {
  console.log("Grade A");
} else {
  console.log("Needs Improvement");
}
```

10. What is it? A shorthand for if...else.

11. Example:

```
let age = 20;
let status = (age >= 18) ? "Adult" : "Minor";
```

12. Nested Ternary

```
let score = 75;  
let result = (score > 90) ? "Excellent" : (score > 60) ? "Good" : "Fail";
```

13. Example:

```
let day = 3;  
switch (day) {  
  case 1:  
    console.log("Monday");  
    break;  
  case 2:  
    console.log("Tuesday");  
    break;  
  default:  
    console.log("Other Day");  
}
```

14. What will be the output?

```
let x = 5;  
let y = "5";  
console.log(x == y);    // ?  
console.log(x === y);   // ?  
console.log(!false);    // ?  
console.log((x > 3 && y < 10) ? "Valid" : "Invalid");
```

15. What is the output of this expression?

```
let a = 3;  
let b = 2;  
let result = a++ + --b;  
console.log(result, a, b);
```

16. Which gives true?

```
null == undefined // ?  
null === undefined // ?
```

17. Login Check

```
let username = "";  
if (username == false) {  
  console.log("Username is empty");  
}
```

18.

```
let user = null;  
let name = user || "Guest";  
console.log(name);
```

19.

```
let isActive = true;  
console.log(!isActive);  
console.log(!isActive);
```

20. Clean Grading System

```
let score = 92;  
if (score >= 90) console.log("🏆 Outstanding");  
else if (score >= 75) console.log("🎯 Excellent");  
else if (score >= 50) console.log("👍 Good");  
else console.log("❌ Fail");
```

21. Can you do multiple conditions?

```
let temp = 40;  
let weather = temp > 35 ? "Hot" : temp < 15 ? "Cold" : "Normal";  
console.log(weather);
```

22.

```
let age = 20;

switch (true) {
  case (age < 13):
    console.log("Child");
    break;
  case (age < 20):
    console.log("Teenager");
    break;
  default:
    console.log("Adult");
}
```

23. Create a function `gradeSystem(score)` that uses:

- if-else OR switch
- And ternary to assign grade like "A", "B" etc.

24. Ask the user's name and age, and print:

- Welcome [name], you are eligible" using ternary

25.

```
let a = 10;
let b = "10";
console.log(a !== b);    // ?
console.log(a !== b);    // ?
console.log(a + ++b);    // ?
```

26. What is the difference between `a = b` and `a == b`?

27. What's the output of `true || false && false`?

28. How do you use the ternary operator for if `age >= 18`?

29.

```
let x = 10;  
console.log(x = x + 1); // OUTPUT?   
console.log(x == x + 1); // OUTPUT? 
```

30.

```
console.log(false == 0);  
console.log(false === 0);
```


31. Form Validation

```
if (username && password) {  
    console.log("Login Allowed");  
} else {  
    console.log("Missing fields");  
}
```

32. Use logical operators for short-circuiting:

33.

```
let isOnline = true;  
let status = isOnline ? "● Online" : "● Offline";
```

34 . Missing Breaks = Bug!

```
let color = "red";  
switch (color) {  
    case "red":  
        console.log("Red");  
    case "blue":  
        console.log("Blue");  
}
```

35. create a Grade Calculator

- Input: Marks (0-100)
- Output: Grade (A+, A, B, etc.)
- Logic: Use if-else or switch

36. create Login Access

- Input: isLoggedIn, isAdmin
- Output: Message using &&, ||, ternary

37. Day to Weekday

- Input: Number (1-7)
- Output: Weekday using switch-case

38. Add 5% tax to a price using +=

39. Check if user is adult using if and age >= 18

40. Use ternary to show "Available" or "Out of Stock"

41. Write switch for subscription type: "Free", "Pro", "Enterprise"

42. Use === to verify secure login (check both value and type)

43. Create grading system using if-else if

44. Create a UI logic: isDarkMode ? " 🌙 " : " ☀ "

45. Use switch(true) for performance band display

46. Add user loyalty points only if isLoggedIn && hasOrderHistory

47. Difference between == and ===? Why is === safer?

48. Can you use switch to compare ranges like if-else?

49. Why do we prefer ternary only for short decisions?

50. How does short-circuiting help with default values?

51. Difference between x++ and ++x?

52. What happens if you forget break in switch?

53. When would you use switch(true) in real-world code?

54. How does JavaScript handle 0, "", null, and undefined in logical checks?

55. fix it

```
let role = "admin";
switch (role) {
  case "admin":
    console.log("Admin");
  case "user":
    console.log("User");
  default:
    console.log("Guest");
}
```

56. What will be the output?

```
let result = 5 > 4 > 3;
console.log(result);
```

57. What's the result of this expression?

58.

```
console.log(null || 0 || "Hello");
```

59. What does this print?

```
console.log(3 && "JS" && 0 && "Rocks");
```

60. Code:

```
let output = (true && false) || (false && true);  
console.log(output);
```

61. Write a one-liner JS code to print "Eligible" if age is 18+ else "Not Eligible".

62. What is the output of this?

```
let a = 10;  
a += 5;  
console.log(a);
```

63.

```
let a = 10;  
a += 5;  
console.log(a);
```

64. What will x++ return if x = 4?

65. What will ++x return if x = 4?

66. What is the result of 10 / 4 in JavaScript?

67. What is the output of:

```
let x = 5;  
x *= 2;  
console.log(x);
```

68. True or False: $x = x - 3$ is same as $x -= 3$?

69. Fill in the blank to increment count by 1:

70. What will be the output?

```
let x = 10;  
let y = x++ + ++x;  
console.log(y);
```

71. What is $3 \% 2$?

72. Predict the result:

```
let a = 2;  
let b = 3;  
let c = a++ + b++ + ++a + ++b;  
console.log(c);
```

73. What's the output of this?

```
let x = 7;  
let y = 3;  
console.log(x % y);
```

74. How do you subtract 4 from a variable num in shorthand?

75. What's the result of 5 ** 2?

76. What is the output of this?

```
let a = 2;  
a *= a + 3;  
console.log(a);
```

77. What happens if you do `a = a + "5"` and `a = 10`?

78. Which operator gives remainder?

79. What's the difference between `85 x++` and `++x`?

80. What is `0 / 5`? And `5 / 0`?

81. True or False: $x ** 3$ is same as $x = x ** 3$?

82. What is the output of:

83. What is the output? Why?

```
console.log(5 == "5");
```

84. What is the output?

```
console.log(5 === "5");
```

85. What is `null == undefined`?

86. What is `null === undefined`?

87. What will this return?

```
console.log(0 == false);
```


88. What will this return?

```
console.log(0 === false);
```

89. What does != mean in JavaScript?

90. What does !== mean in JavaScript?

91. What is the output of:

```
console.log("2" > 1);
```

92. What is the output?

```
console.log("apple" < "banana");
```

93. What is the output?

```
console.log(true == 1);
```

94. What is the output?

```
console.log(false === 0);
```

95. What is the result of:

```
console.log(3 > 2 > 1);
```

96. Which comparison returns true?

```
console.log("5" != 5);      // ?  
console.log("5" !== 5);     // ?
```

97. What is the result?

```
console.log(0 || "JS");
```

98. What is the result?

```
console.log(1 && "React");
```

99.

```
console.log(1 && "React");
```

100. What does ! do in JavaScript?

101. What will this output?

```
console.log(!0);
```

102. What will this output?

```
console.log(!"");
```

103. What will !!"Hello" return?

104. What does this do?

```
let name = null;  
let display = name || "Guest";
```

105. What is short-circuiting in JS?

106. What is the result?

```
false && true && "JS"
```

107. What is the output?

```
console.log("JS" && 0 && true);
```

108. What is the basic syntax of an if statement?

109. What is the output?

```
let age = 20;  
if (age >= 18) {  
    console.log("Adult");  
}
```

110. What is the role of else?

111. What is else if used for?

112. Output?

```
let temp = 30;  
if (temp > 35) {  
  console.log("Hot");  
} else {  
  console.log("Normal");  
}
```

113. Which one is true?

```
let score = 75;  
if (score > 90) {  
  console.log("A");  
} else if (score > 70) {  
  console.log("B");  
} else {  
  console.log("C");  
}
```

114. What will this log?

```
if ("0") {  
  console.log("Yes");  
} else {  
  console.log("No");  
}
```

115. What's wrong here?

```
if (x = 10) {  
  console.log("Works");  
}
```

116. Which values are falsy in if condition?

117. Is this correct?

```
if (false) console.log("No");  
else console.log("Yes");
```

118. Can if work without else?

119. Can else come without if?

120. What happens if no condition is true in if-else if-else chain?

121. What's the benefit of else if over multiple ifs?

122. Can we nest if inside if?

123. Output?

```
let a = 10;  
if (a > 5)  
  if (a < 15)  
    console.log("Perfect");  
else  
  console.log("Too Big");
```

124. What does this print?

```
if (false)  
  if (true)  
    console.log("Hi");  
else  
  console.log("Bye");
```

125. Real-world use case of if-else?

126. Can you use if with logical operators like &&, ||?

127. What is the syntax of the ternary operator?

128. What will this print?

```
let age = 20;  
console.log(age >= 18 ? "Adult" : "Minor");
```

129. Is ternary operator a replacement for if-else?

130. What is the output?

```
let isOnline = false;  
let status = isOnline ? "● Online" : "● Offline";  
console.log(status);
```

131. What happens if condition is true in ternary?

132. Can you nest ternary operators?

133. Output?

```
let x = 5;  
let result = x > 10 ? "Big" : x > 3 ? "Medium" : "Small";  
console.log(result);
```


135. What's wrong here?

```
true ? console.log("Yes") : console.log("No");
```

136. What does this return?

```
false ? 1 : 2;
```

137. Can you assign a value using ternary?

```
let msg = isLoggedIn ? "Welcome" : "Please login";
```

138. Replace with ternary:

```
if (score > 50) msg = "Pass"; else msg = "Fail";
```

139. Output?

```
let x = null;  
let val = x ? "Yes" : "No";  
console.log(val);
```

140. Is this valid?

```
isLoggedIn ? "Yes" : isAdmin ? "Kinda" : "No";
```

141. What are ternary operator alternatives?

142. What is the purpose of break in switch?

143. What happens if you forget break?

144. Output?

```
let day = 1;
switch (day) {
  case 1:
    console.log("Mon");
  case 2:
    console.log("Tue");
  default:
    console.log("Other");
}
```

145. What is the role of default?

146. Can switch work with strings?

147. Output?

```
let color = "blue";
switch (color) {
  case "red":
    console.log("Red");
    break;
  case "blue":
    console.log("Blue");
    break;
}
```

148. What does this print?

```
let role = "admin";
switch (role) {
  case "admin":
  case "superadmin":
    console.log("Access granted");
    break;
  default:
    console.log("Denied");
}
```

149. Can switch be used for range conditions like $x > 5$?

150. Can multiple cases point to same block?

```
case "a":  
case "A":  
  console.log("Letter A");
```

151. What happens if no default is present and nothing matches?

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